

Maximum likelihood estimation for simple linear regression

- Given a simple linear regression model with independent observations.
- $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \sim N(0, \sigma^2), i = 1, \dots, n,$
- the maximum likelihood estimates of β_0 , β_1 , and σ^2 are given by
- $$\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}$$
- $$\widehat{\beta}_1 = \frac{s_{xy}}{s_x^2}$$
- $$\widehat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i)^2$$
- Where $\bar{x}$, $\bar{y}$ are the sample means,
- s_x^2 is sample variance of x
- s_{xy} is the sample covariance between x and y

Maximum likelihood estimation for simple linear regression

- Proof

$$\begin{aligned} p(y|\beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^n p(y_i|\beta_0, \beta_1, \sigma^2) \\ &= \prod_{i=1}^n \mathcal{N}(y_i; \beta_0 + \beta_1 x_i, \sigma^2) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right] \\ &= \frac{1}{\sqrt{(2\pi\sigma^2)^n}} \cdot \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right] \end{aligned}$$

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- and the log-likelihood function

$$\begin{aligned} \text{LL}(\beta_0, \beta_1, \sigma^2) &= \log p(y|\beta_0, \beta_1, \sigma^2) \\ &= -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 . \end{aligned}$$

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$$\frac{dLL(\beta_0, \beta_1, \sigma^2)}{d\beta_0} = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)$$

and setting this derivative to zero gives the MLE for β_0 :

$$\frac{dLL(\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2)}{d\beta_0} = 0$$

$$0 = \frac{1}{\hat{\sigma}^2} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$$

$$0 = \sum_{i=1}^n y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^n x_i$$

$$\hat{\beta}_0 = \frac{1}{n} \sum_{i=1}^n y_i - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n x_i$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} .$$

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The derivative of the log-likelihood function (4) at $\hat{\beta}_0$ with respect to β_1 is

$$\frac{dLL(\hat{\beta}_0, \beta_1, \sigma^2)}{d\beta_1} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i y_i - \hat{\beta}_0 x_i - \beta_1 x_i^2)$$

and setting this derivative to zero gives the MLE for β_1 :

$$\frac{dLL(\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2)}{d\beta_1} = 0$$

$$0 = \frac{1}{\hat{\sigma}^2} \sum_{i=1}^n (x_i y_i - \hat{\beta}_0 x_i - \hat{\beta}_1 x_i^2)$$

$$0 = \sum_{i=1}^n x_i y_i - \hat{\beta}_0 \sum_{i=1}^n x_i - \hat{\beta}_1 \sum_{i=1}^n x_i^2$$

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$$0 \stackrel{(6)}{=} \sum_{i=1}^n x_i y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) \sum_{i=1}^n x_i - \hat{\beta}_1 \sum_{i=1}^n x_i^2$$

$$0 = \sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i + \hat{\beta}_1 \bar{x} \sum_{i=1}^n x_i - \hat{\beta}_1 \sum_{i=1}^n x_i^2$$

$$0 = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} + \hat{\beta}_1 n\bar{x}^2 - \hat{\beta}_1 \sum_{i=1}^n x_i^2$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - \sum_{i=1}^n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - \sum_{i=1}^n \bar{x}^2}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} .$$

Maximum likelihood estimation for simple linear regression

The derivative of the log-likelihood function (4) at $(\hat{\beta}_0, \hat{\beta}_1)$ with respect to σ^2 is

$$\frac{dLL(\hat{\beta}_0, \hat{\beta}_1, \sigma^2)}{d\sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

and setting this derivative to zero gives the MLE for σ^2 :

$$\frac{dLL(\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2)}{d\sigma^2} = 0$$

$$0 = -\frac{n}{2\hat{\sigma}^2} + \frac{1}{2(\hat{\sigma}^2)^2} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\frac{n}{2\hat{\sigma}^2} = \frac{1}{2(\hat{\sigma}^2)^2} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 .$$

The Stochastic Gradient Descent Algorithm

- correct value is $y = 1$ (this is a positive review),
- and with a feature vector $x = [x_1, x_2]$ consisting of these two features:
 $x_1 = 3$ (count of positive lexicon words)
- $x_2 = 2$ (count of negative lexicon words)
- Let's assume the initial weights and bias in θ^0 are all set to 0, and the initial learning rate η is 0.1:
- $w_1 = w_2 = b = 0$
- $\eta = 0.1$

The Stochastic Gradient Descent Algorithm

The single update step requires that we compute the gradient, multiplied by the learning rate

$$\theta^{t+1} = \theta^t - \eta \nabla_{\theta} L(f(x^{(i)}; \theta), y^{(i)})$$

In our mini example there are three parameters, so the gradient vector has 3 dimensions, for w_1 , w_2 , and b . We can compute the first gradient as follows:

$$\nabla_{w,b} L = \begin{bmatrix} \frac{\partial L_{\text{CE}}(\hat{y}, y)}{\partial w_1} \\ \frac{\partial L_{\text{CE}}(\hat{y}, y)}{\partial w_2} \\ \frac{\partial L_{\text{CE}}(\hat{y}, y)}{\partial b} \end{bmatrix} = \begin{bmatrix} (\sigma(\mathbf{w} \cdot \mathbf{x} + b) - y)x_1 \\ (\sigma(\mathbf{w} \cdot \mathbf{x} + b) - y)x_2 \\ \sigma(\mathbf{w} \cdot \mathbf{x} + b) - y \end{bmatrix} = \begin{bmatrix} (\sigma(0) - 1)x_1 \\ (\sigma(0) - 1)x_2 \\ \sigma(0) - 1 \end{bmatrix} = \begin{bmatrix} -0.5x_1 \\ -0.5x_2 \\ -0.5 \end{bmatrix} = \begin{bmatrix} -1.5 \\ -1.0 \\ -0.5 \end{bmatrix}$$

The Stochastic Gradient Descent Algorithm

Now that we have a gradient, we compute the new parameter vector θ^1 by moving θ^0 in the opposite direction from the gradient:

$$\theta^1 = \begin{bmatrix} w_1 \\ w_2 \\ b \end{bmatrix} - \eta \begin{bmatrix} -1.5 \\ -1.0 \\ -0.5 \end{bmatrix} = \begin{bmatrix} .15 \\ .1 \\ .05 \end{bmatrix}$$

So after one step of gradient descent, the weights have shifted to be: $w_1 = .15$, $w_2 = .1$, and $b = .05$.

References

- <https://statproofbook.github.io/P/slr-mle.html>
- <https://web.stanford.edu/~jurafsky/slp3/5.pdf>